

Tutorial 3 exercise

Part 1: ArrayList with Strings

Q1 - Shopping List

Create an `ArrayList<String>` called `shoppingList` and:

1. Add the items: "Milk", "Bread", "Eggs", "Butter"
2. Print the entire list.
3. Print only the item at index 2.
4. Insert "Coffee" at index 1 (between Milk and Bread).
5. Remove "Eggs" from the list (remove by value, not index).
6. Print the size of the list.
7. Check whether "Bread" is in the list using `contains()` and print the result.
8. Print each item on its own line using a for-each loop.
9. Print each item with its index using a classic for loop (e.g., 0: Milk).
10. Add a new item Espresso in the middle of the ArrayList
(size/2 to get the middle index)
11. Replace every item with its UPPERCASE version using `set()`.
12. Remove all items that are longer than 5 characters.

Part 2: ArrayList with Numbers

Q2 — Exam Scores

Create an `ArrayList<Integer>` called `scores` containing: 78, 92, 45, 88, 61, 99, 53

1. Print the highest and lowest score (loop manually - don't use Java function).
2. Calculate and print the average score (watch out for integer division!).
3. Count how many students passed (score ≥ 50).
4. Create a new `ArrayList<Integer>` containing only the passing scores.
5. Sort the scores in ascending order using `Collections.sort()`.
6. Sort them in descending order. (Hint: `Collections.reverseOrder()`)
7. Add 5 more score 88, then remove all occurrences of 88.

Part 3: ArrayList with Objects (Person)

First, create this class:

```
public class Person {
    private String name;
    private int age;
    private String city;

    public Person(String name, int age, String city) {
        this.name = name;
        this.age = age;
        this.city = city;
    }

    public String getName() { return name; }
    public int getAge() { return age; }
    public String getCity() { return city; }
    public void setAge(int age) { this.age = age; }

    @Override
    public String toString() {
        return name + " (" + age + ", " + city + ")";
    }
}
```

Q3 - People Directory

Create an ArrayList<Person> and add:

Name	Age	City
Alice	30	Singapore
Ben	25	Beijing
Cara	26	Singapore
Dev	30	Seoul

1. Print all people.
2. Print only the names of people living in Singapore.
3. Find and print the oldest person.
4. Increase everyone's age by 1 (use the setter).
5. Remove all people younger than 28.

Part 4: HashMap

Q4 - Friend Email (HashMap<String, String>)

1. Create a map of friend → email with at least 5 entries.
2. Print the email of one specific friend using get().

What happens when you get() a friend that isn't in the map?

Use getOrDefault() to return "Unknown" for missing countries.

3. Update the email of one friend using put(), what happens to the old value?
4. Remove one friend's entry and print the final map size.

Q5 - Student Grades (HashMap<String, Integer>)

Create a map: "Alice" → 85, "Ben" → 72, "Cara" → 90, "Dev" → 65

1. Print every student and grade using a for-each loop over entrySet().
2. Print only the names using keySet().
3. Calculate the average grade using values().
4. Check whether "Cara" exists with containsKey().
5. Add 5 bonus points to every student's grade. (Hint: loop over entrySet() and use entry.setValue(...).)

Q6- Word Frequency Counter

Given the string:

String text = "the quick brown fox jumps over the lazy dog the fox";

Count how many times each word appears and store the result in a HashMap<String, Integer>.

Expected output (order may vary!):

the=3, fox=2, quick=1, brown=1, jumps=1, over=1, lazy=1, dog=1

Part 5: HashMap with Objects

Q7 - Employee Registry (HashMap<String, Person>)

Use the Person class from above.

Create a map where the key is an employee ID (e.g., "E001") and the value is a Person.

1. Add 4 employees.
2. Look up and print the person with ID "E003".
3. Print all IDs and the corresponding person's name (e.g., E001 -> Alice).
4. Given an ID, increase that person's age by 1.
5. What happens if you put() a new Person with an existing ID? Try it and explain.

Q8 - Grouping: City → List of People (HashMap<String, ArrayList<Person>>)

This combines both collections!

1. Loop over an ArrayList<Person> and build a map grouping people by city.
2. Print each city and the people who live there.

Expected structure:

Singapore -> [Alice (30, Singapore), Cara (26, Singapore)]

Beijing -> [Ben (25, Beijing)]

Seoul -> [Dev (30, Seoul)]